Project Documentation

Project Title

Visualization Tool for Electric Vehicle Charge and Range Analysis

1. Objective

The objective of this project is to analyze Electric Vehicle (EV) data and create interactive Tableau dashboards that help users understand EV distribution, charging range, vehicle types, and brand-wise trends.

2. Problem Statement

The increasing adoption of electric vehicles has generated large amounts of data. It is difficult to identify trends and compare EV performance using raw datasets. This project transforms the data into interactive visualizations for better decision-making.

3. Dataset

Source: Kaggle

Dataset: Electric Vehicle Population Data

Fields Used:

Make

Model

Model Year

Electric Vehicle Type

Electric Range

Base MSRP

City

State

County

4. Tools Used

Tableau Desktop 2026.2

GitHub

Microsoft Excel

5. Data Preprocessing

Imported dataset into Tableau.

Removed unnecessary fields.

Checked for missing values.

Created calculated fields.

Applied filters.

6. Calculated Fields

Range Category

IF [Electric Range] >= 200 THEN "High Range"

ELSE "Low Range"

END

Vehicle Count

COUNT([Model])

7. Visualizations Created

Top 10 EV Brands

EV Distribution by Model Year

EV Type Distribution

EV Distribution by State

Average Electric Range by Brand

Base MSRP by Brand

Top Cities by EV Count

Dashboard with Filters

8. Dashboard

The dashboard combines multiple charts into one interactive interface with filters for vehicle brand and EV type.

9. Story

A Tableau Story was created to present the analysis step by step and explain important insights.

10. Performance Testing

Dashboard loads successfully.

Filters work correctly.

Charts update dynamically.

Story navigation works properly.

11. Results

Tesla has the highest number of EVs.

Battery Electric Vehicles are more common than Plug-in Hybrid EVs.

Some brands offer significantly higher electric ranges.

EV adoption has increased in recent years.

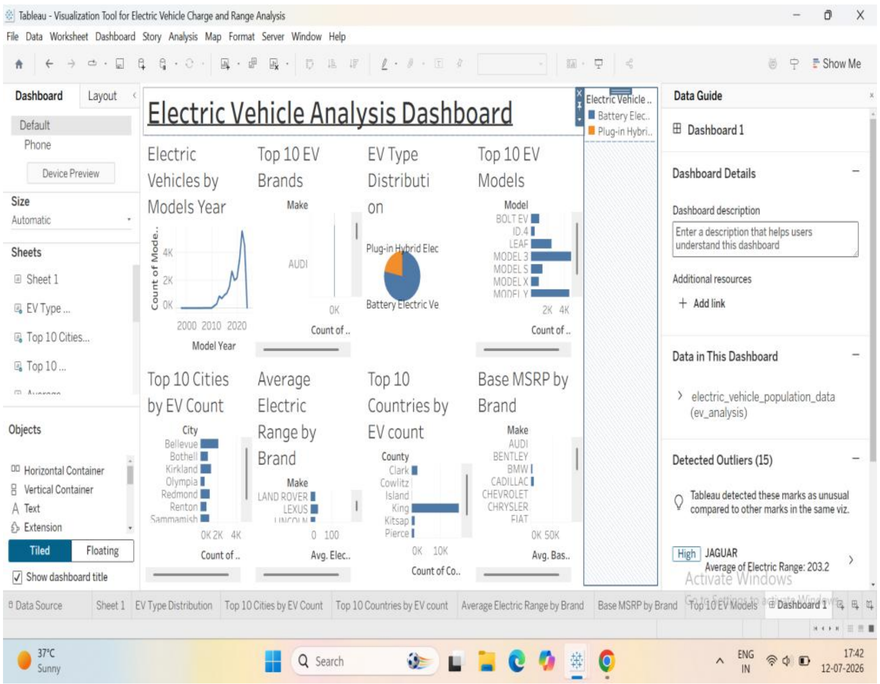
12. Conclusion

The project successfully converts raw EV data into meaningful visualizations using Tableau. The dashboard helps users analyze EV trends and supports better understanding of electric vehicle adoption.

13. GitHub Repository

https://github.com/Poojitha-chitti/Visualization-Tool-for-Electric-Vehicle-Charge-and-Range-Analysis

14.Screenshots of dashboard and story



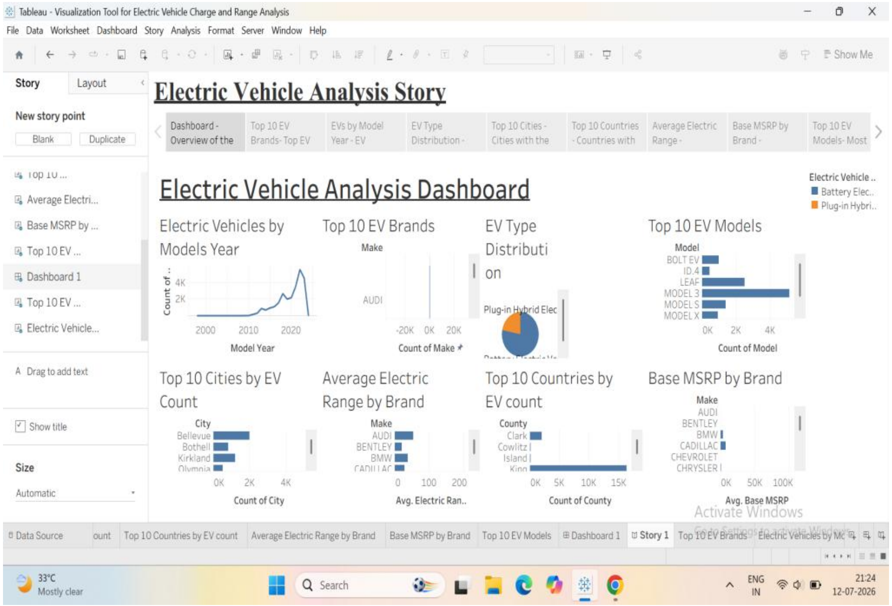


Tableau - Visualization Tool for Electric Vehicle Charge and Range Analysis
Electric Vehicle Analysis Story
Electric Vehicle Analysis Dashboard
Electric Vehicles by Models Year
Top 10 EV Brands
EV Type Distribution
Top 10 EV Models
Top 10 Cities by EV Count
Average Electric Range by Brand
Top 10 Countries by EV count
Base MSRP by Brand